Do Now · retrieval
Entity relationship diagrams

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Where is a FK stored in a one-to-many relationship?

Q2

What does PK stand for?

Q3

What is an associative entity?

Q4

A BST root20 has left10 and right30; insert 25. Describe comparisons and placement.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — ERDs model entities and relationship cardinalities before implementing tables.

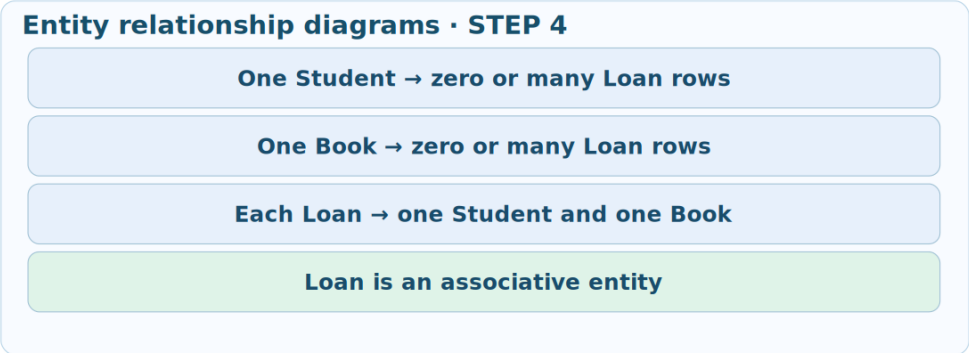
Core idea 2

Apply — One-to-many relationships are typically represented with a foreign key on the many side.

Core idea 3

Evaluate — Many-to-many relationships require an associative/junction table when translated into a relational schema, often with two foreign keys.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

ERD — Entity relationship diagram.

cardinality — Number of related instances.

junction table — Table linking M:N records.

one-to-many — One entity instance can have several related rows.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

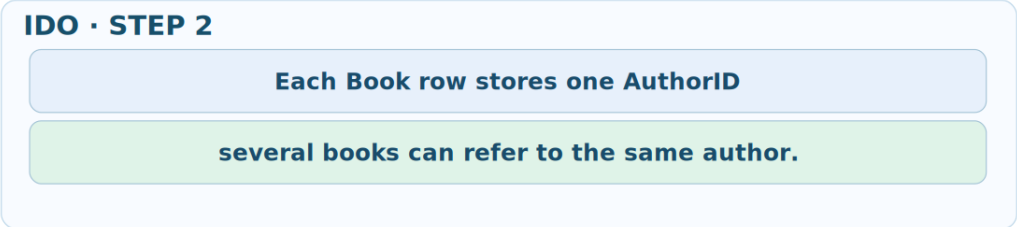
Problem

Model Author and Book when one author writes multiple books, each book has one primary author.

Follow the worked method

1. Predict first — Each Book row stores one AuthorID
2. Model answer — several books can refer to the same author.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first
Create Enrolment junction table

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

State meanings of 1:1,1:M and M:N cardinalities. [3]

Question 2

Draw/describe an ERD for Customer and Order with one customer placing many orders. Give foreign key. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Resolve Student many-to-many Course into relational tables; describe keys and integrity. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A many-to-many relationship is usually represented in relational tables by placing one simple FK on either table.

Use key vocabulary from memory

1. Explain ERD in your own words.

2. Explain cardinality in your own words.

3. Explain junction table in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.